# Literature status: the tight-fitting conjecture in arXiv:2502.08406

Answer completed by arXiv:2607.20055v2 and its cited sharp-contraction results

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**Status: fully answered in later literature.** Conjecture 22 of Bao–Ma–Yan–Zhu is now a consequence of Kalaj's proof of the Gromov–Ros conjecture, arXiv:2607.20055v2, together with the source paper's own embedding and compactness theorems. Kalaj's Corollary 7.2 gives the sharp Bergman contraction and all equality cases; Corollary 7.3 gives the Hardy endpoint. A one-line intermediate-norm argument below also gives all Hardy endpoint equality cases.

## 1 Question and normalization dictionary

Conjecture 22 of arXiv:2502.08406 asks whether, for $a, b > -1$, the only tight fittings on $\mathbb{B}^n \subset \mathbb{C}^n$ are

$$\begin{aligned} A^p_a \subset A^q_b: &\quad p < q, \qquad (n+1+a)q = (n+1+b)p, \\ H^p \subset A^q_a: &\quad p < q, \qquad nq = (n+1+a)p, \end{aligned}$$

and whether equality of the two norms occurs exactly for

$$\begin{aligned} f(z) &= \frac{c}{(1-\langle z, w\rangle)^{2(n+1+a)/p}} \qquad &&\text{in the first line,} \\ f(z) &= \frac{c}{(1-\langle z, w\rangle)^{2n/p}} &&\text{in the second line,} \end{aligned}$$

where $c \in \mathbb{C}$ and $w \in \mathbb{B}^n$.

Kalaj uses invariant measure

$$d\mu_n(z) = \frac{dv(z)}{(1-|z|^2)^{n+1}}$$

and writes $\mathcal{A}^p_A$ for the normalized norm with integrand $|f|^p(1-|z|^2)^A \, d\mu_n$, where $A > n$. Thus

$$A = n+1+a, \qquad B = n+1+b, \qquad \mathcal{A}^p_A = A^p_a, \qquad \mathcal{A}^q_B = A^q_b.$$

In this notation the first critical line is $q/p = B/A$ and the second is $q/p = A/n$.

## 2 Later theorem and the two contractions

Li and Su [2] had proved the required higher-dimensional convex contraction conditionally on geodesic balls being all isoperimetric regions in complex hyperbolic space. Kalaj [3] proves exactly this missing input in Theorem 6.9 for every $n \geq 2$. His Theorem 7.1 therefore makes the sharp convex Bergman contraction unconditional. In particular, Corollary 7.2 says that for $A, B > n$ and $q/p = B/A > 1$,

$$\|f\|_{\mathcal{A}^q_B} \leq \|f\|_{\mathcal{A}^p_A},$$

with equality, for nonzero $f$, exactly for

$$f(z) = c(1 - \langle z, w\rangle)^{-2A/p}.$$

The one-dimensional inequality is Kulikov's theorem; the strict-convexity equality principle cited by Kalaj (Nicola–Riccardi–Tilli, Theorem 5.2) gives the same equality family. This proves the analytic assertion in part (a) of Conjecture 22 in every dimension.

For the Hardy endpoint, put $r = p/n$ and $A = n + 1 + a$. The critical identity $nq = Ap$ is exactly $q = rA$. Kalaj's Corollary 7.3, the now-unconditional Hardy endpoint of the Li–Su chain, gives

$$\|f\|_{\mathcal{A}_A^{rA}} \le \|f\|_{H^{rn}},$$

which is precisely $\|f\|_{A_a^q} \le \|f\|_{H^p}$. The corresponding disc statement is again contained in Kulikov's one-dimensional result.

# 3 Endpoint equality cases

The full Hardy equality classification follows without any additional geometric argument. Choose an arbitrary $C$ with $n < C < A$. The contractive chain gives

$$\|f\|_{\mathcal{A}_A^{rA}} \le \|f\|_{\mathcal{A}_C^{rC}} \le \|f\|_{H^{rn}}.$$

If the two endpoint norms are equal, then equality holds in the first inequality. The Bergman equality statement just quoted, applied with source parameters $(C, rC)$ and target parameters $(A, rA)$, forces

$$f(z) = c(1 - \langle z, w\rangle)^{-2C/(rC)} = c(1 - \langle z, w\rangle)^{-2/r} = c(1 - \langle z, w\rangle)^{-2n/p}.$$

Conversely these kernels are extremal, as already observed in the source paper (or by ball-automorphism invariance). The case $f = 0$ is included by taking $c = 0$. This proves exactly the extremal assertion in part (b).

# 4 Why this proves "if and only if" for tight fittings

It remains only to combine the sharp inequalities with results already proved in arXiv:2502.08406. Its Theorems A–C show that a noncompact proper embedding with $a, b > -1$ can occur in the two displayed families only on the stated critical line and with $p < q$. Conversely, on that critical line the embedding is proper and noncompact. The later sharp inequalities make it contractive; constants show that its operator norm is exactly one. Hence it is a tight fitting. This proves both directions and both equality classifications of Conjecture 22.

# 5 Verification note

The source question was checked in Conjecture 22 of arXiv:2502.08406. The later status was checked against arXiv:2607.20055v2 (revised 7 August 2026), especially Theorem 6.9, Theorem 7.1, Corollary 7.2, and Corollary 7.3, and against the conditional theorem and parameter convention in arXiv:2411.01911. The parameter shift $A = n + 1 + a$ is forced by changing from Euclidean weighted volume to invariant volume, and makes both critical lines identical term by term.

## References


[1] G. Bao, P. Ma, F. Yan, and K. Zhu, *Embedding and compact embedding between Bergman and Hardy spaces*, arXiv:2502.08406; J. Geom. Anal. 36 (2026).

[2] X. Li and G. Su, *Contraction property on complex hyperbolic ball*, arXiv:2411.01911, 2024/2025.

[3] D. Kalaj, *The Gromov–Ros conjecture in complex hyperbolic space*, arXiv:2607.20055v2, 2026.

[4] F. Nicola, F. Riccardi, and P. Tilli, *The Wehrl-type entropy conjecture for symmetric $SU(N)$ coherent states: cases of equality and stability*, arXiv:2412.10940, especially Section 5.